Graph Cardinality Preserved Attention Network for Fault Diagnosis of Induction Motor Under Varying Speed and Load Condition

Yao Tang, Xiaofei Zhang, Guojun Qin, Zhuo Long, Shoudao Huang, *Member, IEEE*, Dianyi Song, and Haidong Shao

***Abstract*—During the long-term operation of motors, their working conditions are changing due to the industrial demands or declining health status, and traditional diagnosis methods perform poorly in that case. This article proposes a fault diagnosis method based on graph cardinality preserved attention network (GCPAT), which can work under varying working conditions, and can be generalized to the transient state. Diagnosis results are obtained by analyzing signal-converting graphs, which are composed of nodes and edges. First, the vibration signals are converted into symmetrical snowflake images by symmetrized dot pattern (SDP) method. Second, SLIC is developed to make homogeneous super-pixels in SDP images as nodes, and form graphs according to color, texture, and distance features. Finally, the GCPAT is utilized to distinguish motor status. Compared with other state-of-art methods, the results show the out-performance of GCPAT under varying working conditions both in steady and transient state.**

***Index Terms*—Fault diagnosis, graph neural network (GNN), rotating motors, symmetrized dot pattern (SDP).**

I. Introduction

ELECTRIC motors are significant components of Industry 4.0 era (electrification, digitization, and interconnectivity) due to its complex physical operating environment and varying working condition [1], [2]. During long-term operation, these machines are possible to suffer from various faults, which would adversely affect the manufacturing productivity, even lead to large economic costs and terrible catastrophes [3]. Therefore, accurate fault diagnosis algorithm for electric motor is of great importance.

Due to the rapid development of computer science, intelligent methods have attained remarkable achievement in fault diagnosis. They can learn discriminative features automatically, and do not rely on the professional knowledge [4]. However, most of the existing intelligent methods only work under limited working condition [5]–[7], which are unsatisfied with the practical industrial demands. Recently, to tackle this issue, several intelligent fault diagnosis methods have been developed focusing on time or frequency domain. A multibranch and multiscale convolutional neural network (CNN) method was proposed for diagnosis under varying load conditions [8]. A domain adaptive CNN was developed to obtain strong domain-invariant capacity [9]. The intelligent network was designed which can be generalized to varying working conditions [10]. Although these methods show certain advantages, because the traditional time-frequency methods are strict to working environment, they are still inherent with the following weaknesses.

1) These methods only focus on the conditions that load or speed changes in a small degree.
2) They would fail when load or speed is in the transient state.
3) These preprocessing approaches strongly depend on professional knowledge.

Recently, an intuitive signal representation method, the symmetrized dot pattern (SDP) has been employed in fault classification [11], [12]. Unlike these conventional methods, SDP can directly transform original signals into mirrored symmetrical snowflake images with easy implementation and small computational burden [13]. The recognition and discrimination between patterns are directly related to the amplitude and frequency differences between signals. Its robustness against noise has been verified in [14], and a few researches have applied SDP in tougher working environment. Wang proposed the bearing fault (BF) detection method when load change in wide range [15]. Gu combined the SDP and angular domain resampling technology to realize diagnosis when speed is in transient state [16]. Accordingly, the SDP method is effective and reliable for fault diagnosis. Its application potential under varying working conditions and transient state has been demonstrated and needs further study.

Manuscript received February 20, 2021; revised June 4, 2021 and July 9, 2021; accepted September 12, 2021. Date of publication September 16, 2021; date of current version February 18, 2022. This work was supported in part by the National Science Foundation of China under Grant 52077064 and in part by the Foundation of Key Laboratory of Science and Technology on Integrated Logistics Support under Grant 6142003200203. Paper no. TII-21-0754. *(Corresponding authors: Xiaofei Zhang; Guojun Qin.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TII.2021.3112696.

Digital Object Identifier 10.1109/TII.2021.3112696

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Due to the SDP method, fault diagnosis tasks transform to image classification tasks, and CNNs have been widely utilized in this field [17]. Generally, CNNs process images in Euclidean domain and only focus on the value in matrix. Composed by nodes and edge connections, graphs provide an informative data structure, which contains the information including values and the connected relationships between the internal data. Recently, graph neural networks (GNNs) have revealed its potential in the fault diagnosis [18], [19]. In general, GNNs realize classification in nodes, edges, or graphs level. Zhang diagnosed the BF based on node classification [20]. In order to construct a graph, features of each image are extracted and are aggregated into nodes. Therefore, nodes represent samples and node label indicates the fault type. But in this way, the overall features of image are presented but the details are ignored.

In GNNs, graph attention networks (GATs) [21] have been widely studied and applied. GATs show the superiority by assigning adaptive attention weights to different connected nodes. Zhang [22] verified the ability of cardinality preservation to improve the performance in the attention mechanism. Therefore, a fault diagnosis method based on graph attention cardinality preservation network (GACPT) is proposed, which can work under varying working conditions. First, the raw vibration signals are transformed into the SDP images. Second, the simple linear iterative cluster (SLIC) is developed to group pixels adaptively into homogeneous super-pixels to form nodes, and the edge connections are decided by nodes spatial distances. Thereby, the detailed features in image can be noticed and fault diagnosis is achieved in graph level instead of node level. Finally, the experimental results for five motor statuses under varying working conditions verified that the graph cardinality preserved attention network (GCPAT) achieves the superior performance compared with other baseline methods. The main contributions of our article are concluded as follows.

1) The SDP method is studied with signals in steady and transient state. Due to the consistency of images, this article reveals that SDP is an effective signal-to-image method for fault diagnosis when load or speed is changing.
2) The proposed fault diagnosis method can analyze constructed graphs and realize accurately diagnosis under varying working conditions.

The rest of this article is organized as follows. The theoretical backgrounds of above methods are presented in Section II. A detailed description of the proposed method is given in Section III. In Section IV, experiment setup is explained and the superiority of proposed method is verified by comparing with other intelligent algorithms. Finally, Section V concludes this article.

II. Theoretical Background

A. SDP Method

The SDP method is a signal-to-image method. Based on the kth and the $(k + l)$ th values in the discrete time series, this algorithm establishes the mapping relationship between Cartesian coordinate and polar coordinate, presenting signals in the form of symmetrical snowflake-shaped figures.

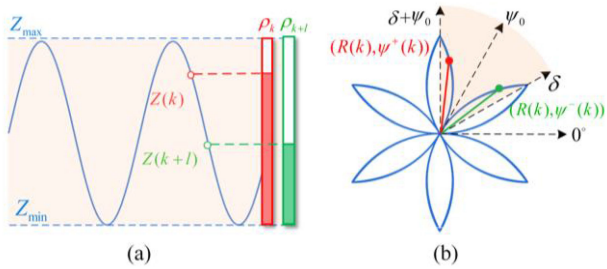


Fig. 1. Illustration of SDP method's principle. (a) Time series. (b) SDP image.

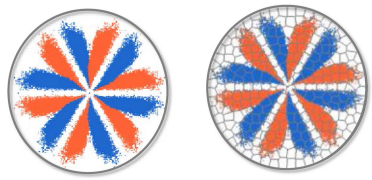


Fig. 2. Image segmented by SLIC into about 200 super-pixels.

Fig. 1 takes sine series $Z(i)$ to clarify the transformation. First, it is shown that the periodic curve is bounded in range from $Z_{\min}$ to $Z_{\max}$. By defining $\triangle(i) = Z(i) - Z_{\min}$, $\triangle(k)$ and $\triangle(k+1)$ are easily obtained and $\triangle_{\max} = Z_{\max} - Z_{\min}$. Then, through $\rho(i) = \triangle(i)/\triangle_{\max}$, $\rho(k)$ and $\rho(k+l)$ are calculated for forming radius and angle in polar space based on (1), where k is the kth value in original time signals, l represents the time lag coefficient; ψ_0 is the initial rotation angle; δ is the rotation range. Generally, ψ_0 is chosen as 60°, indicating six symmetrical petals; δ ranges from 0° to 60°and l is between 1 to 10 [23]

$$\begin{cases} R(k) = \rho(k) \\ \psi^+(k) = \psi_0 + \delta * \rho(k+l) \\ \psi^-(k) = \psi_0 - \delta * \rho(k+l) \end{cases}. \tag{1}$$

Thereby, $Z(k)$ and $Z(k+l)$ are converted to $(R(k),\psi^+(k))$ and $(R(k),\psi^-(k))$. It is known that $R(k)\in(01)$, $\psi^+(k)\in(\psi_0,\delta+\psi_0)$, and $\psi^-(k)\in(\delta,\psi_0)$, so the mapped angle range is within ψ_0, and the remaining dots in SDP are positioned symmetrically.

B. SLIC Method

The SLIC algorithm is an adaptation of k-means to group pixels into expressively atomic regions, so-called super-pixels, aiming to downgrade the complexity of image processing by substituting the rigid pixel grid structure, as shown in Fig. 2.

By default, the only parameter needed to be set is: n, indicating the desired number of divided super-pixels. First, image I with N pixels is converted to CIELAB. It is a color space developed by international commission on illumination, where u indicates luminance, a and b are combined to represent color. Then, based on position information x and y, cluster centers $C_i = [u_i, a_i, b_i, x_i, y_i]$ are sampled on image I, which is divided into roughly equally super-pixels with size $S = (N/k)^{1/2}$. According to lowest gradient position, the centers are shifted to seed locations to

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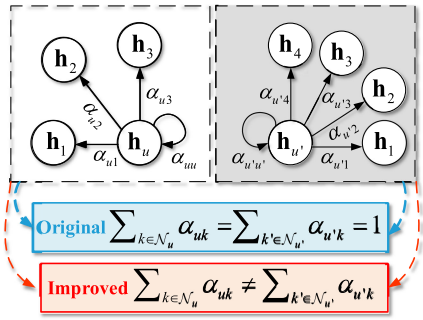


Fig. 3. Explanation of original and improved attention mechanism in aggregators on two distinct multiset with different node distribution.

avoid error clustering at the image edge. Finally, the assignment step and update step are needed to associate pixels with nearest cluster center and the final cluster centers are adjusted [24].

C. GCPAT

GNNs analyze graphs by exchanging neighborhood information recurrently to learn the embedding representation, so called neighbor aggregation and graph pooling. As one type of GNNs, GATs assign trainable weights in aggregation to distinguish the information between a target node and its neighbor nodes. However, the traditional GATs ignore the certain distinct connect relationships of nodes, which limits the discriminative power of the algorithm. Therefore, the cardinality information is needed to improve aggregation [22].

Fig. 3 is utilized to explain the principle of cardinality preservation. For source node u and u', the edge connections are different. In the original algorithm, regardless of the neighbor structure of the node, the sum of attention weights of one node remains at 1. However, in cardinality preservation algorithm, the sum of attention weights is related to the number of edges, so the neighbor relationship can be reflected. Therefore, the summation function of attention weight is modified and the cardinality information is utilized to update node representation during aggregation.

The hidden representation $\boldsymbol{h}l\ u$ of the node u is

$$\boldsymbol{h}_u^l = f^l\left(\psi^l\left(|\mathcal{N}_u|\right)\odot\sum_{v\in\mathcal{N}_u}\alpha_{uv}^{l-1}\boldsymbol{h}_v^{l-1}\right) \tag{2}$$

where the subscript l indicates l-th layer, and $\alpha l\text{-}1\ uv$ is the attention weight between node u and its connected node v. N_u is a node set contains all the neighbors. $\boldsymbol{h}l\text{-}1\ v$ represents hidden state of node v in $(l-1)$th layer. $|N_u|$ indicates the cardinality information of node u. Through nonlinear function $\psi^l(\cdot)$ and $f^l(\cdot)$, the hidden state of target node can be updated.

III. Framework of the Proposed Fault Diagnosis Method

The overall framework of the proposed method is illustrated in Fig. 4, including: signal-image-graph method and end-to-end networks. The details would be described below.

A. Signal-Image-Graph Method

Step 1: Obtain the original time-series data. The radial vibration signals are measured from motors with different status under varying speed and load working conditions.

Step 2: Form visual distinguished SDP patterns. One dimensional time-series are converted to snowflake images. In the SDP method, because the rotation range d and time lag coefficient l will affect the image presentation, based on the principle of maximizing image difference, images similarities are calculated as Fig. 5, and parameters are chosen as $\delta = 30$ and $l = 4$.

Step 3: SLIC is performed as an intermediate step from image to graph. It adaptively divides an image into smaller super-pixels. The only parameter needed to decide is the desired number of super-pixels n. In our article, n is set as 200 to ensure an affordable computation and the details of the image can be described.

Step 4: Construct graphs based on super-pixels. The basic information contained in a graph includes node features and edge connections. First, the color and texture features are treated as node features. The color features are composed of l, a, and b in CIELAB color space, and texture features include mean and standard deviation value of angular second moment, entropy, contrast, correlation, and inverse different moment. They are five typical representative figures in gray-level co-occurrence matrix. Hence, the node feature consists of three color features and ten texture features. Then, The spatial and color distance are combined to determine the comprehensive distance between nodes [25]. A node is connected to the five closest nodes in the article.

B. Architecture of GCPAT

As listed in Table I, the architecture of the proposed network contains two parts: GCPAT and multilayer perception (MLP). After four convolutional-pooling blocks, outputs out_1–out_4 are obtained, and the sum of them is utilized as the input of MLP, which is used to produce the final diagnosis results of motor status. In the Table I, the number of nodes in all graphs is denoted by *nn*. The number of node attributes is denoted by *na*. The hidden layer size is represented by *nh*, which is set as 32. The number of fault types is *nf*.

IV. Experimental Verification

In this section, the diagnostic performance of GCPAT under steady and transient state is demonstrated by comparing with the intelligent methods.

A. Implementation Details

1) Data Description: From the test rig as Fig. 6 and induction motor as Table II, the radial vibration data are measured including five statuses: 1) healthy condition (HC); 2) BBF; 3) BF; 4) ITSF, and 5) RBF. The details of fault implantation are

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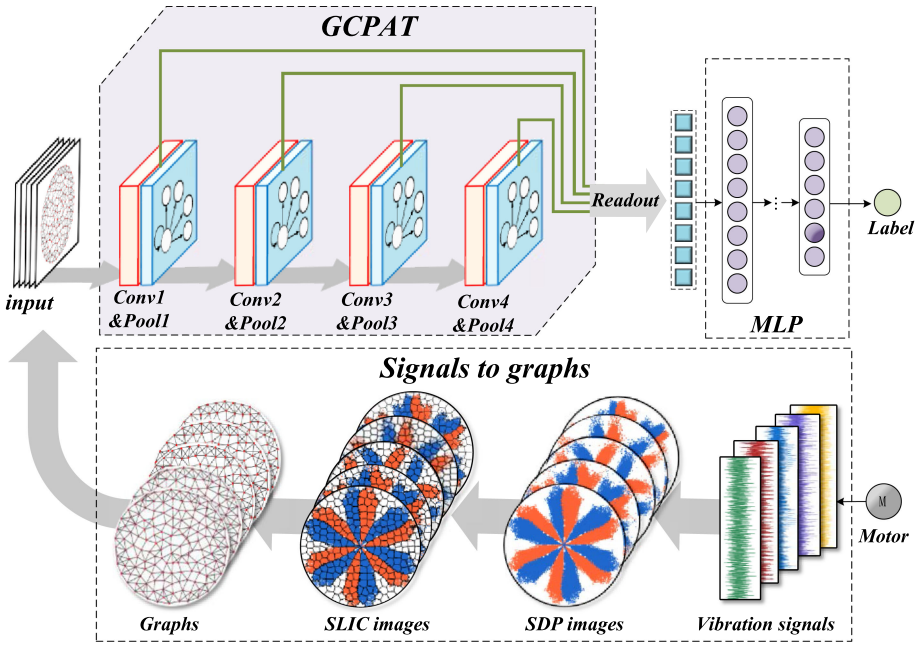


Fig. 4. Framework of the proposed fault diagnosis method based on GCPAT.

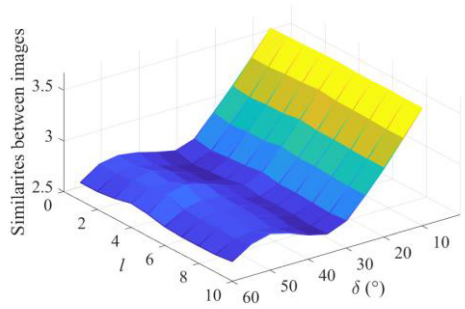


Fig. 5. Relationship between δ, l, and images similarities.

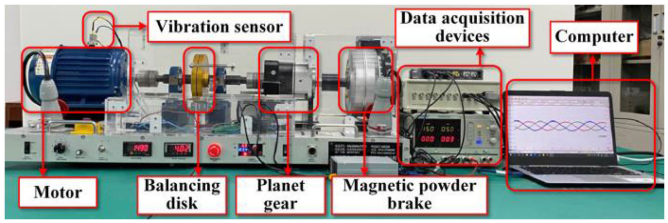


Fig. 6. Overviews of the test rig.

TABLE I
STRUCTURE AND PARAMETERS OF THE GCPAT

	Layer	Type	Input Dimension	Activation	Output Dimension
GCPAT	$Conv_1$	Convolution	x_0: [*nn*,*na*]	ReLu	x_1: [*nn*,*nh*]
	$MaxPool_1$	MaxPooling	x_1: [*nn*,*na*]	/	out_1: [1,*nh*]
	$Conv_2$	Convolution	x_1: [*nn*,*na*]	ReLu	x_2: [*nn*,*nh*]
	$MaxPool_2$	MaxPooling	x_2: [*nn*,*na*]	/	out_2: [1,*nh*]
	$Conv_3$	Convolution	x_2: [*nn*,*na*]	ReLu	x_3: [*nn*,*nh*]
	$MaxPool_3$	MaxPooling	x_3: [*nn*,*na*]	/	out_3: [1,*nh*]
	$Conv_4$	Convolution	x_3: [*nn*,*na*]	ReLu	x_4: [*nn*,*nh*]
	$MaxPool_4$	MaxPooling	x_4: [*nn*,*na*]	/	out_4: [1,*nh*]
MLP	Input	Sum	out_1: [1,*nh*] out_2: [1,*nh*] out_3: [1,*nh*] out_4: [1,*nh*]	/	x_5: [1,*nh*]
	Fc_1	Linear	x_5: [1,*nh*]	ReLu	x_6: [1,*nh*]
	Fc_2	Linear	x_6: [1,*nh*]	ReLu	x_7: [1,*nh*]
	Fc_3	Linear	x_7: [1,*nh*]	ReLu	x_8: [1,*nh*]
	Fc_4	Linear	x_8: [1,*nh*]	ReLu	x_9: [1,*nh*]
	Fc_5	Linear	x_9: [1,*nh*]	ReLu	out: [1,*nf*]

TABLE II
RATED PARAMETERS OF TESTED INDUCTION MOTOR

Poles	Voltage	Current	Frequency	Speed	Power
4	220/380V	6.4/3.7A	50Hz	1500rpm	2hp

TABLE III
PARAMETERS OF VIBRATION SENSOR

Type	Voltage sensitivity	Frequency range	Resolution ratio
CT1000LA	5mV	0.5~10kHz	0.02g

shown in Fig. 7. For BBF, 4 of 28 rotor bars are cut out, and the motor imbalance is adjusted by compensating mass. For BF, the inner BF is equipped in the fan end, and the outer BF is equipped in the drive end. For ITSF, a coil of the stator winding is led out, and short circuit fault is determined by a switch. The short circuit ratio is about 1/3. For RBF, the bending ratio of the shaft at both ends is about 40/100 mm. These data were sampled by 5 kHz and each sample contains 10k points. The parameters of the vibration sensor are listed in Table III.

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TABLE IV
SAMPLE DESCRIPTION OF DATA-SET *A*, *B*, AND *C* IN STEADY STATE

Motor status	Size of data-set A/B/C								
	speed=1000rpm			speed=1250rpm			speed=1500rpm		
	0 hp	1 hp	2 hp	0 hp	1 hp	2 hp	0 hp	1 hp	2 hp
HC	34/28/23	33/28/22	33/28/22	34/28/23	33/28/22	33/27/22	34/28/22	33/28/22	33/27/22
BBF	34/28/23	33/28/22	33/28/22	34/28/23	33/28/22	33/27/22	34/28/22	33/28/22	33/27/22
BF	34/28/23	33/28/22	33/28/22	34/28/23	33/28/22	33/27/22	34/28/22	33/28/22	33/27/22
ITSF	34/28/23	33/28/22	33/28/22	34/28/23	33/28/22	33/27/22	34/28/22	33/28/22	33/27/22
RBF	34/28/23	33/28/22	33/28/22	34/28/23	33/28/22	33/27/22	34/28/22	33/28/22	33/27/22
Total	1500/1250/1000								

TABLE V
SAMPLE DESCRIPTION OF DATA-SETS *D* AND *E* WHEN LOAD IS IN TRANSIENT STATE

Motor status	Size of data-set D_1 (1000rpm) / D_2 (1250rpm) / D_3 (1500rpm)				Size of data-set E_1 (1000rpm) / E_2 (1250rpm) / E_3 (1500rpm)			
	0~0.5hp	0.5~1.0hp	1.0~1.5hp	1.5~2.0hp	2.0~1.5hp	1.5~1.0hp	1.0~0.5hp	0.5~0.0hp
HC	13	13	12	12	13	13	12	12
BBF	13	13	12	12	13	13	12	12
BF	13	13	12	12	13	13	12	12
ITSF	13	13	12	12	13	13	12	12
RBF	13	13	12	12	13	13	12	12
Total	250/250/250				250/250/250			

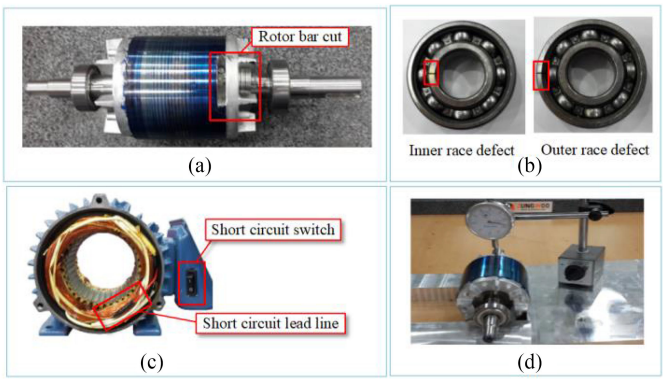


Fig. 7. Fault implantation diagram of (a) BBF, (b) BF, (c) ITSF, and (d) RBF.

The data-sets in Table IV are utilized to study the impact of the sample size on the algorithms. The numbers of samples in each working condition are described. The speed includes 1000, 1250, and 1500 r/min, and the load includes 0, 1, and 2 hp, so there are nine working conditions in total. Based on the experiment, there are 200 samples in each working condition, and the samples in the data-sets are random selected. The whole samples numbers of data-sets *A*, *B*, and *C* are 1500, 1250, and 1000, respectively. Each data-set is divided in to three parts for training, validating and testing, and the proportions of each part are 60%, 20%, and 20%.

The data-sets in Tables V and VI are developed to test whether the models trained by steady-state data-sets are robust to generalize to transient state. Data-sets D_1, D_2, and D_3 are obtained when the load is rising with the speed of 1000, 1250, and 1500 r/min, respectively, and E_1, E_2, and E_3 are produced when the load is falling. Data-sets F_1, F_2, and F_3 are obtained when the speed is rising with the load of 0, 1, and 2 hp respectively, and G_1, G_2, and G_3 are produced when the speed is falling. The load and speed are manually adjusted with a relative stable rate, about 0.25 hp/s for load and 50 r/min/s for speed. These data-sets are used for testing with the number of 250.

2) Baseline Methods: To verify the effectiveness of the proposed fault diagnosis method, the following intelligent algorithms are developed as comparisons, including: GAT [26], visual geometry group network (VGG-16) [27], hierarchical graph pooling with structure learning (HGP-SL) [28], LeNet-5 [29], multilayer perception (MLP) [30] and long short-term memory [31]. The hyper-parameters including learning rate and batch size are selected by grid search approach. The inputs of GNN methods (GAT and HGP-SL) are graphs, and for other methods, the SDP images are performed as inputs. These methods are performed on a computer with Intel Core: Intel(R) Xeon(R) Silver 4214 CPU, RAM: 128GB, and NVIDIA GeForce RTX 2060 GPU. For the proposed method, the required diagnostic time for a sample is about 14.28 ms.

B. Experimental Results in Steady State

1) SDP Images for Data-Sets in Steady State: Taking steady state data-sets under different load conditions at 1000 r/min as examples, the time series and SDP images are shown in Fig. 8. It can be seen that the differences between different motor statuses are distinguishable. Although the load varies from no load to full load, the changes between the SDP images in the same motor status are not significant.

Fig. 9 shows images under different speed at full load. It can be seen that the huge interclass inconsistency is presented. For example, in HC status, the patterns of different speed are inconsistent from each other. The great outer-class ambiguity is also introduced. For instances, the images of HC and BBF are

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TABLE VI
SAMPLE DESCRIPTION OF DATA-SETS *F* AND *G* WHEN SPEED IS IN TRANSIENT STATE

Motor status	Size of data-set F_1 (0 hp) / F_2 (1 hp) / F_3 (2 hp)					Size of data-set G_1 (0 hp) / G_1 (2 hp) / G_3 (2 hp)				
	1000~1100 rpm	1100~1200 rpm	1200~1300 rpm	1300~1400 rpm	1400~1500 rpm	1500~1400 rpm	1400~1300 rpm	1300~1200 rpm	1200~1100 rpm	1100~1000 rpm
HC	10	10	10	10	10	10	10	10	10	10
BBF	10	10	10	10	10	10	10	10	10	10
BF	10	10	10	10	10	10	10	10	10	10
ITSF	10	10	10	10	10	10	10	10	10	10
RBF	10	10	10	10	10	10	10	10	10	10
Total	250/250/250					250/250/250				

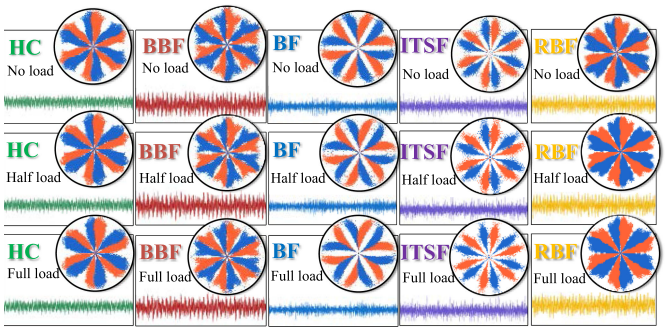


Fig. 8. Vibration signals and SDP images of HC, BBF, BF, ITSF, and RBF motors under different load conditions with the speed of 1000 r/min.

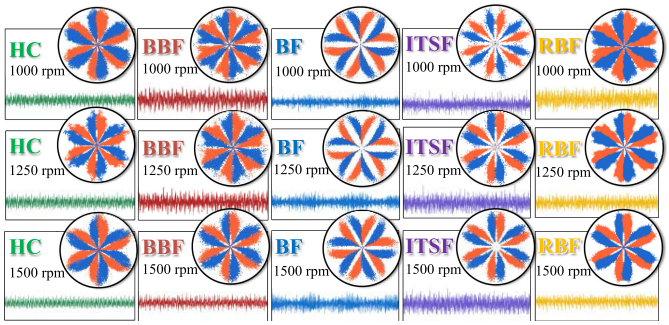


Fig. 9. Vibration signals and SDP images of HC, BBF, BF, ITSF, and RBF motors under different speed conditions with the load of 2 hp.

highly similar at 1500 r/min. Hence, the above-mentioned issues enhance the difficulty in accurate fault diagnosis.

2) Comparison With Baseline Methods: Receiver operating characteristic (ROC) curves are effective to compare the performance of algorithm on each motor statue. TP, FP, TN, and FN are denoted as the number of true positive, false positive, true negative, and false negative, correspondingly. The horizontal axis and the vertical axis of the ROC curves are false positive rate and true positive rate respectively, which are calculated as

$$\text{False positive rate (FPR)} = \text{FP}/(\text{FP} + \text{TN}) \tag{3}$$

$$\text{True positive rate (TPR)} = \text{TP}/(\text{TP} + \text{FN}). \tag{4}$$

ROC curves of five motor statues and their microaverage and macroaverage are shown in the Fig. 10. Macroaverages of FPR and TPR are the average of FPR and TRP over five classes. To obtain microaverages of FPR and TPR, first, TP, FP, TN, and FN of each class are summed, then it is calculated by the (3) and (4) based on the summation variables. The area under the curve (AUC) is utilized to measure the quality of the classification, which ranges from 0.5 to 1. The AUC value of 1 indicates a perfect fault diagnosis without the misdiagnosis. In Fig. 10, AUV values are listed in parenthesis. It is shown that the curves of GCPAT based on *A*, *B*, and *C* data-sets are closest to the upper left corner, with the maximum AUC values of 0.9998, 0.9996, and 0.9991 for microaverage, and of 0.9998, 0.9997, and 0.9994 for macroaverage. The overall performance of the proposed method is the superior compared with the baseline methods. Also, it achieves high AUC values on five motor statues. Among these methods, the classification of BF achieves best performance compared with other four motor statues. When the number of sample decreases, as data-set *A* to data-set *C*, it is obvious that the AUC values of GCPAT are still close to 1, whereas the values of baseline methods are decreased to some extent. Therefore, it can be concluded that the proposed method realizes effective fault diagnosis.

Based on data-set B, T-distributed stochastic neighbor embedding (t-SNE) [32] is presented in Fig. 11. It can be observed that via deep learning methods, the features become distinguishable to some extent, and GCPAT can completely separate these features. Among all methods, the features of BF are best separated. Also, the confusion matrices on data-set *B* are shown as Fig. 12. The confusion matrix is a 5×5 matrix, in which each row and column represents the motor statuses, and the horizontal and vertical axes represent the predicted and true labels, respectively. The proportions (bold) and the number of classifications are denoted in the confusion matrix. It is clear that the GCPAT reaches the highest diagnostic accuracy.

C. Experimental Results in Transient State: Load Changed and Speed Fixed

1) SDP Images for Transient State Data-Sets: Fig. 13(a) and (b) show the SDP images of HC motors when load continuous rises or falls at a rate about 0.25 hp/s. The speed condition is initially set at 1000 r/min, and it would not be adjusted deliberately. However, with the change of load it would also change in a small degree. The signals take 8 s and divide them into four parts to generate corresponding SDP images. It should be noticed that the transient and steady state images are generated sequentially based on time series, with intervals of 2 s, therefore the number of

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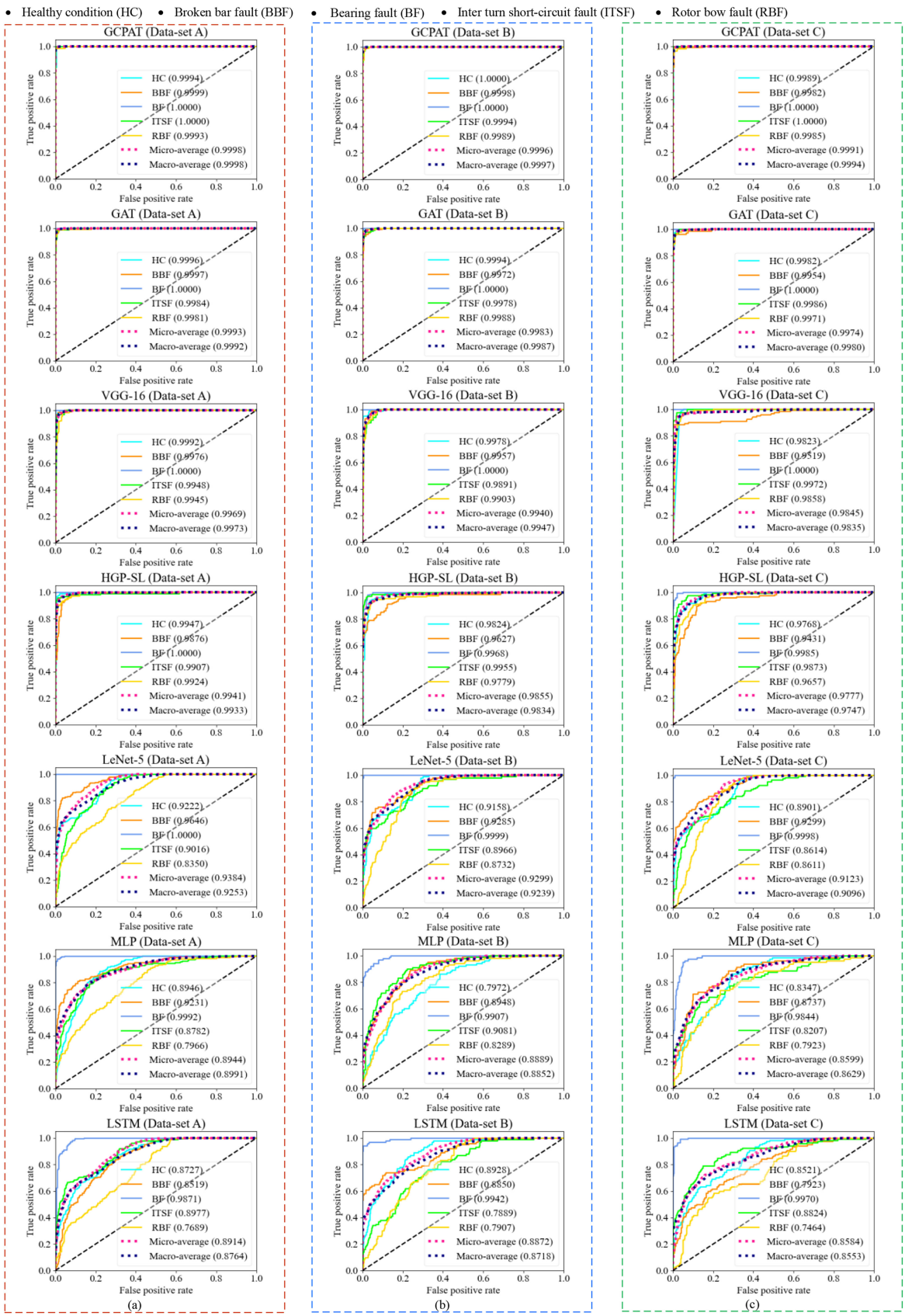


Fig. 10. ROC plots via deep learning methods based on (a) data-set *A*, (b) data-set *B*, and (c) data-set *C*.

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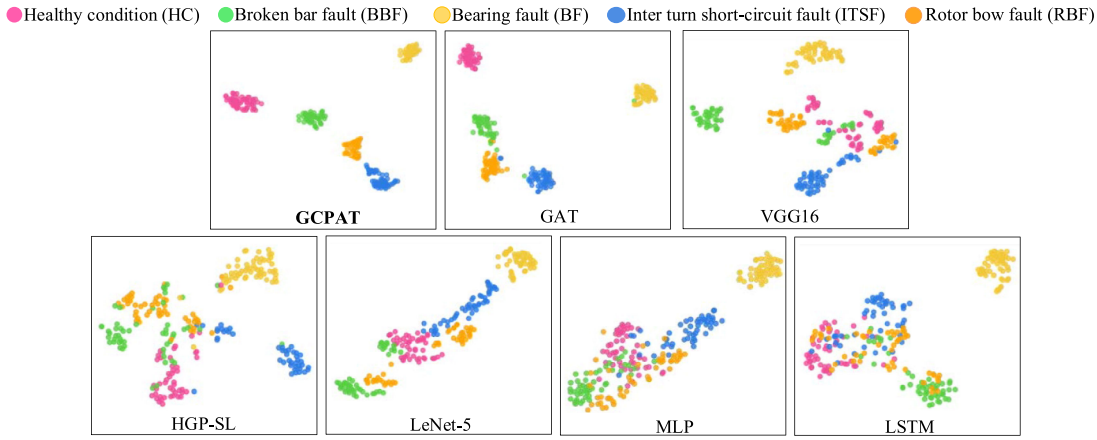


Fig. 11. 2-D t-SNE visualization of features of data-set *B* via deep learning methods.

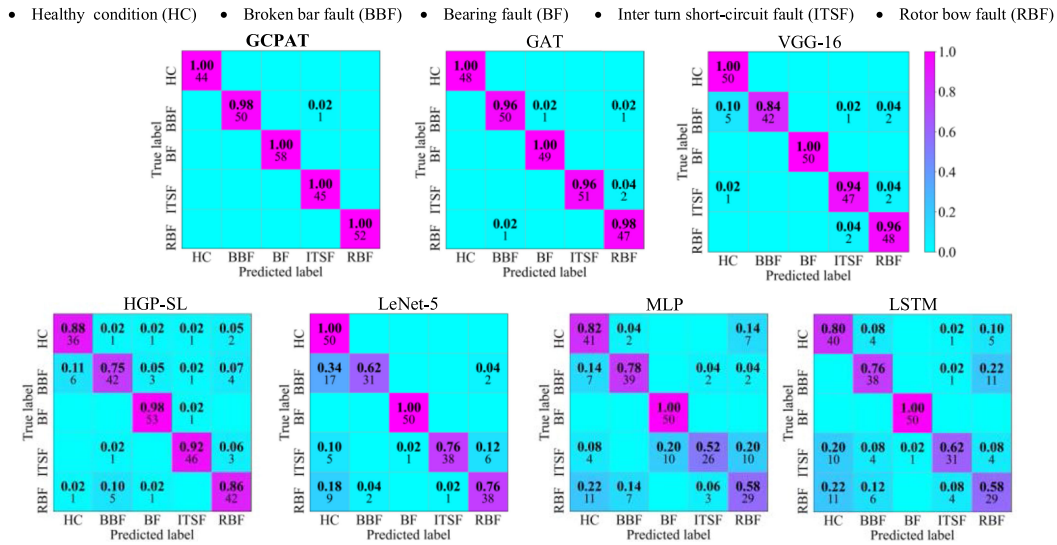


Fig. 12. Confusion matrices of data-set *B* via deep learning methods.

points in each transient sample should be identical with trained stable sample. Compared with the SDP images in Fig. 13(c), it can be observed that when the load changes within a certain range, it has little effect on the SDP image. In other words, the images in transient state and steady state are highly consistent. Therefore, the SDP method can be applied to diagnosis when load is changing.

The SDP images in Fig. 13(a) and (b) are random selected as HC samples in D_1, E_1 respectively, and BF, BBF, ITSF, and RBF samples are produced in the same way. The data-set D_2 E_2 and D_3 E_3 contain the samples with the speed of 1250 and 1500 r/min, respectively. The detail information of data-sets is listed in Table V.

2) Comparison With Baseline Methods: Based on models trained by data-sets *C*, the ROC plots (microaverage) for different methods based on data-sets *D* and *E* are shown in Fig. 14. For data-set D, the GCPAT reaches the highest AUC values of 0.9960, 0.9950, and 0.9917 for D_1, D_2, and D_3, respectively. For data-set E, the GCPAT obtains the highest AUC values of 0.9987, 0.9987, and 0.9927 for E_1, E_2, and E_3, respectively. It is demonstrated that the proposed method trained by samples in steady state can make the diagnosis when load is changing in the corresponding range. The results prove that the proposed method has the satisfying generalization ability in load transient state.

D. Experimental Results in Transient State: Speed Changed and Load Fixed

1) SDP Images for Transient State Data-Sets: Fig. 15 shows SDP images of BBF motor when load fixed at 2 hp, and speed continuous rises or falls at a rate about 50 r/min/s. The signals cost 10 s and are divided into five parts to generate corresponding SDP images. Also, the number of points of each sample should stay the same with steady samples. Compared Fig. 15(a) and (b) with (c), it can be found that images change with rotating

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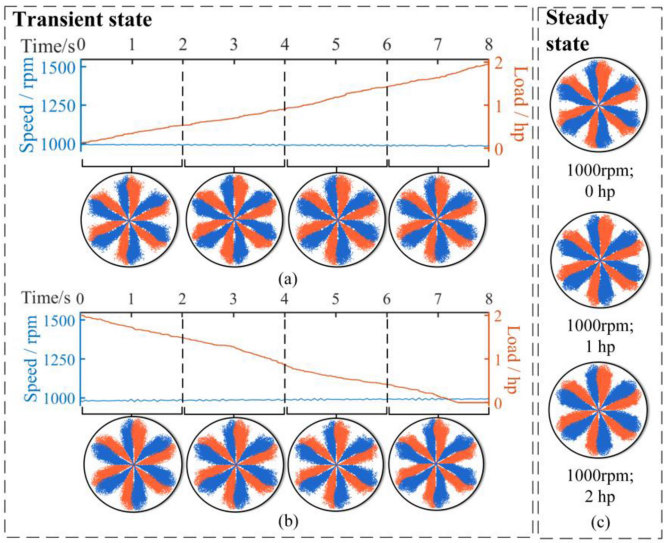


Fig. 13. SDP images of HC motors when speed is fixed at 1000 r/min. (a) Load is increased from 0 to 2 hp. (b) Load is decreased from 2 to 0 hp. (c) Load is fixed.

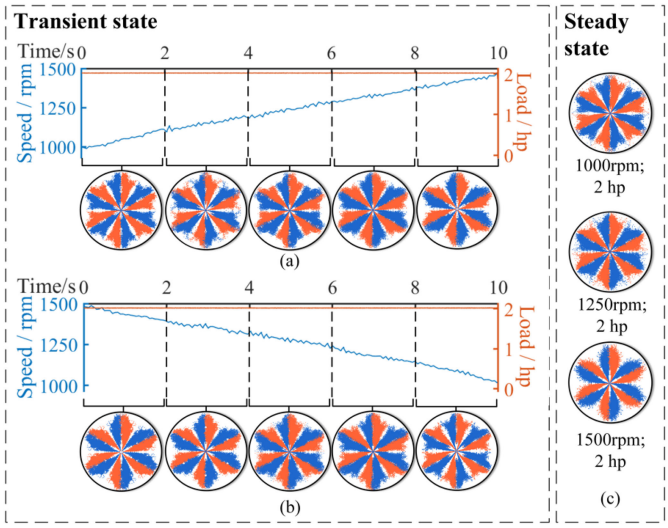


Fig. 15. SDP images of BBF motor when load is fixed at 2 hp. (a) Speed is increased from 1000 to 1500 r/min. (b) Speed is decreased from 1500 to 1000 r/min. (c) Speed is fixed.

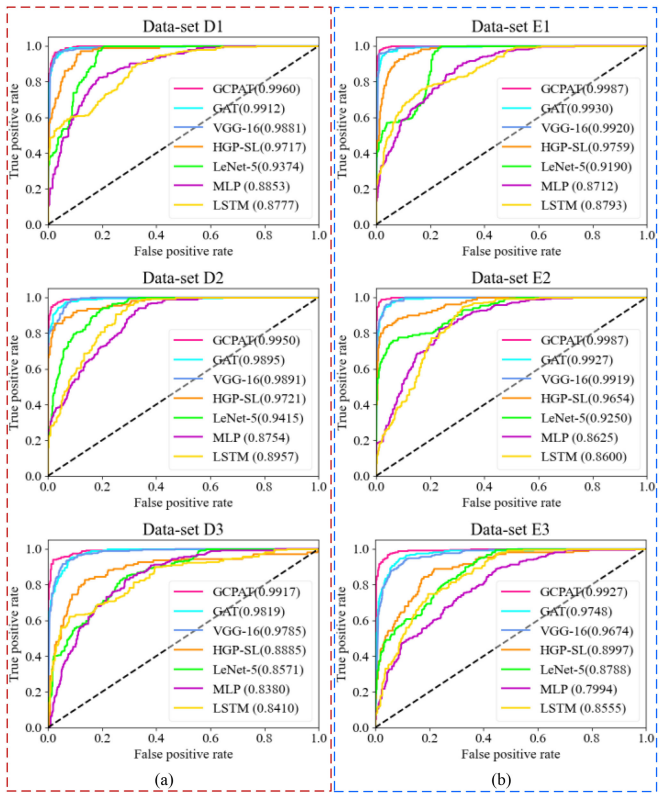


Fig. 14. ROC curves via deep learning methods based on (a) data-set *D* and (b) data-set *E*.

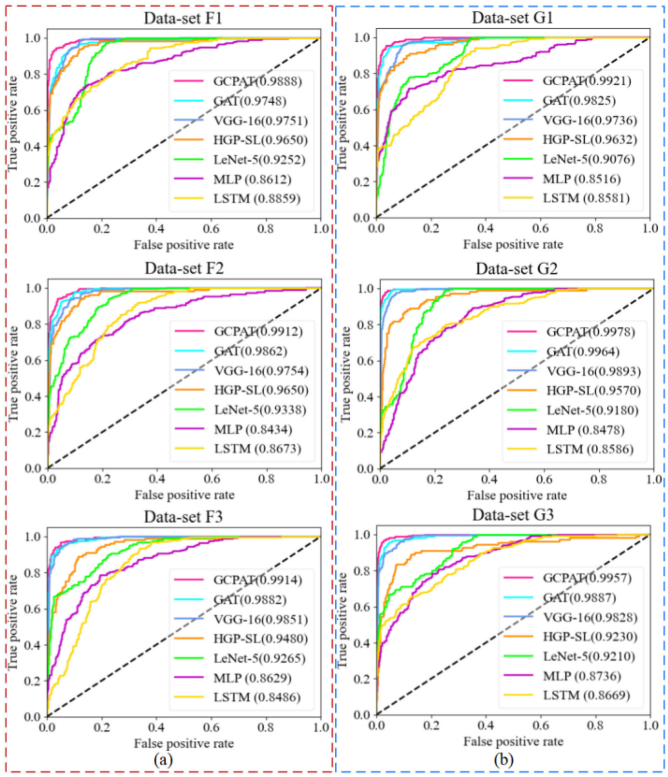


Fig. 16. ROC curves via deep learning methods based on (a) data-set F and (b) data-set G.

speed. When the speed is changing, the SDP images generated in a certain range of speed is similar to the steady-state SDP images which are close to that range, and it is effective both for the rising and the falling curves. Thereby, the conclusion can be drawn that the SDP method is an effective signal-to-image method for diagnosis when speed is in transient state.

The SDP images formed from Fig. 15(a) and (b) are random selected as BBF samples in F_3, and G_3 respectively, and HC, BF, ITSF, and rotor bow fault (RBF) samples are produced in the same way. The data-set F_1 G_1 and F_2 G_2 contain the samples

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with the load of 0 and 1 hp, respectively. The detail information of data-sets is listed in Table VI.

2) Comparison With Baseline Methods: Based on models trained by data-sets C, the ROC plots (microaverage) for different methods based on data-set F and G are shown in Fig. 16. For data-set F, the GCPAT reaches the highest AUC values of 0.9888, 0.9912, and 0.9914 for F_1, F_2, and F_3 respectively. For data-set G, the GCPAT obtain the highest AUC values of 0.9921, 0.9978, and 0.9957 for G_1, G_2, and G_3, respectively. The results prove that the proposed method trained with samples in steady state can make the diagnosis when speed is changing in the corresponding range. These results prove that the superior generalization ability of GCPAT in speed transient state.

V. Conclusion

In this article, a fault diagnosis method based on GCPAT was proposed for accurate fault diagnosis under steady and transient condition. The proposed method was trained by steady state data, which were graphs formed based on SDP method and SLIC method. The results showed that the SDP images in steady and transient state were similar. The SDP method was effective for fault diagnosis when load or speed was changing. The analysis of experiment results also revealed that the proposed method achieved the strong diagnosis ability in steady state and superior generalization ability in the transient state. The proposed method suggested a promise for real-world industrial applications of motor fault diagnosis.

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